



PROTOTYPE EVALUATION REPORT



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Assets and Content

Asset	Appropriateness	Reliability	Legal and ethical implications
Organic Carrots	This was not appropriate to use to show a picture for the product. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image for the product being sold.	The source was unreliable since it did not actually give the picture of product being sold. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution the real picture for the product will be used.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual product looks like.
Fresh Milk	This was not appropriate to use to show a picture for the product. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image for the product being sold.	The source was unreliable since it did not actually give the picture of product being sold. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution the real picture for the product will be used.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual product looks like.
Seasonal Fruit Jam	This was not appropriate to use to show a picture for the product. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image for the product being sold.	The source was unreliable since it did not actually give the picture of product being sold. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution the real picture for the product will be used.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual product looks like.
sophie	This was not appropriate to use to show a picture for the	The source was unreliable since it did not actually give the	This asset has the legal implication to be used freely except in logos,

	producer. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image of the producer working with Greenfield Local Hub.	picture of the producer selling their product. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution real pictures of producers will be used that are working with Greenfield Local Hub.	trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual producer looks like.
charlotte	This was not appropriate to use to show a picture for the producer. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image of the producer working with Greenfield Local Hub.	The source was unreliable since it did not actually give the picture of the producer selling their product. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution real pictures of producers will be used that are working with Greenfield Local Hub.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual producer looks like.
emily	This was not appropriate to use to show a picture for the producer. Since, the picture is not professional which would be industry standard for other companies. Furthermore, the picture does not show the actual image of the producer working with Greenfield Local Hub.	The source was unreliable since it did not actually give the picture of the producer selling their product. However, this does not matter for the solution as this is a prototype that is not going to be used by the company. In the live solution real pictures of producers will be used that are	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not representing how the actual producer looks like.

		working with Greenfield Local Hub.	
hero_image	This was not appropriate to use to show a picture for the hero image. Since the picture contains a person smoking a cigarette which is not relevant providing a reminder of healthy, locally produced food. Furthermore, the picture was not professional which is expected as it is industry standard.	The source was unreliable since the picture did not represent how the real food market looks like at Greenfield Local Hub.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of not showing how a real market looks like at Greenfield Local Hub.
cta_image	This was not appropriate to use for the call-to-action image. This is because the image is not convincing enough for users to trust buying from the website, which is learned from the feedback that has been gathered. Furthermore, the picture was low-quality and not professional which is not industry standard.	The source was unreliable since the picture did not represent a person that works at or with Greenfield Local Hub.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This has the ethical implication of showing a person on the picture that is not related to Greenfield Local Hub.
shopping-cart	This was appropriate to use, since images like the current image are used to represent a basket/cart for the user. Due these users are less likely to get confused with what the button does, which the image is used for.	The source is reliable since it has been used by other developers as an icon to represent a basket/cart for prototypes. For a live solution a custom icon could be used to fit the website's design more.	This asset has the legal implication to be used freely except in logos, trademarks or to ruin the someone's reputation. This asset has no ethical implication.
Hero section slogan	This was appropriate to use as it fits the duty of telling the users the	The source is reliable since it has been used by a similar	This asset has no legal implication since the original text was modified

	purpose of this solution in a few words.	website that was being actual when the asset was retrieved.	to fit the GLH solution and therefore it will not be plagiarism. This asset has no ethical implication.
'Why buy locally?' information	This was appropriate to use as it fulfils its duty to convince people to buy food from local producers.	The source is not reliable as it was AI, but it was proofread by me to ensure unrelated information is not included.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.
Sophie's details	This was not appropriate to use since they are not the real producer's details that work with Greenfield Local Hub. Also, the details are not detailed enough which will not inform the users fully.	The source is not reliable as it was AI, and it did not take a real producer's details that work with Greenfield Local Hub. Therefore, a reliable source would be needed in the live solution.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.
Emily's details	This was not appropriate to use since they are not the real producer's details that work with Greenfield Local Hub. Also, the details are not detailed enough which will not inform the users fully.	The source is not reliable as it was AI, and it did not take a real producer's details that work with Greenfield Local Hub. Therefore, a reliable source would be needed in the live solution.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.
Charlotte's details	This was not appropriate to use since they are not the real producer's details that work with Greenfield Local Hub. Also, the details are not detailed enough which will not inform the users fully.	The source is not reliable as it was AI, and it did not take a real producer's details that work with Greenfield Local Hub. Therefore, a reliable source would be needed in the live solution.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.
Organic Carrots details	This was not appropriate to use since they are not the real product's details that work with Greenfield Local Hub. Also, the details are not	The source is not reliable as it was AI, and it did not take a real product's details that work with Greenfield Local Hub. Therefore, a	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.

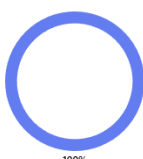
	detailed enough which will not inform the users fully.	reliable source would be needed in the live solution.	
Fresh Milk details	This was not appropriate to use since they are not the real product's details that work with Greenfield Local Hub. Also, the details are not detailed enough which will not inform the users fully.	The source is not reliable as it was AI, and it did not take a real product's details that work with Greenfield Local Hub. Therefore, a reliable source would be needed in the live solution.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.
Seasonal Fruit Jam details	This was not appropriate to use since they are not the real product's details that work with Greenfield Local Hub. Also, the details are not detailed enough which will not inform the users fully.	The source is not reliable as it was AI, and it did not take a real product's details that work with Greenfield Local Hub. Therefore, a reliable source would be needed in the live solution.	This asset has no legal implication since the text was generated by AI and modified to fit the GLH solution.

Project Outcomes

The overall prototype met most of the original requirements for the solution for Greenfield Local Hub. It was only mostly met as some features do not functional or functional differently from the requirements that were proposed initially to the client.

Functional Requirements

Functional Requirement	Met	Explanation and Future Improvement
Users able to register	Yes	This is met fully because users can register accounts easily by either typing in the email and password or using Google to create an account. Normal register (see below) 48. Were you able to create an account by entering in your details? More details  Using Google register (see below)

		49. Were you able to create an account by clicking on the Google button? More details ● Yes 2 ● No 0  In the future, users should have more ways to register such as using a Microsoft account.
Users able to log in	Yes	This is met fully because users can log in to their account that they created using their manually created account or using Google. Normal login 51. Were you able to log in to your account by entering your details? More details ● Yes 2 ● No 0  Using Google login 52. Were you able to log in to your account by clicking on the Google button? More details ● Yes 2 ● No 0  In the future, users should have more ways to log in such as using a Microsoft account.
Users able to change their log in details	Yes	This is met fully as users can navigate to their profile where they can change their email and password. In the future, it could be changed by allowing certain users, like the admin able to change other user's details.
Users able to view privacy information	Yes	This is met fully as there was a complete privacy page containing necessary privacy information for the users understand how their data is being used according to GDPR. In the future, the privacy page should be laid out in a more readable way, since the layout does not match professionally made websites.
Users able to view information about the producers, their methods and the benefits of buying locally	Not fully	This was not met fully since the producers' methods were not outlined on the producers' page. In the future these details will be added on, so customers trust the producers working with Greenfield Local Hub and are more likely to buy products from them.
Customers able to view products with transparent pricing and availability	Not fully	This was no met fully because the actual prices of the products were shown clear on products page, baskets' page and the order page. However, the availability was not shown and

		therefore customers could buy product that was not available. This is shown in the feedback received (see below). 19. How clear were the prices shown on the website? More details - Very clear 0 - Clear 2 - Somewhat unclear 0 - Very unclear 0 100% In the future customers should not be able to see or buy product that is not available.
Customers able to place orders for collection or delivery	Yes	This was met fully because customers can select how they want their order and fill in the relevant information for the order type. However, this could be improved in the future by allowing more order types.
Customers able to view order history	Yes	This was met fully, since customers can access the orders' page where their order history, which shows the number of products they ordered. This is supported by the feedback received (see below). 60. Were you able to view all of your orders you made? More details - Yes 2 - No 0 100% However, this could be improved in the future by allowing a button to see more details of the order clear on a separate page just for a specific product. This improvement will make the user experience better which will increase the likelihood of a user being retained for longer.
Provide a dashboard for producers	Not fully	This was no met fully, since a dashboard was provided but the producers were unable to view or edit stock amount for a product. In the future all functionality for the producer dashboard should be available to use by producers. 72. What would you add to the dashboard page for it to be more helpful at managing products and orders? More details 2 Responses Latest Responses "A page where they can view orders" "A details button, and an edit button to change the status"
Provide order management, scheduling and tracking features	Not fully	Because users cannot change the status of their order through cancelling it a customer or changing the status as an admin or producer. In the future, there should be a button for customers to cancel their orders, and producers should be able to use a dropdown to change an order's status from their dashboard.
A loyalty scheme that provides discounts and specials offers	Not fully	Since there are discounts which any customer can use despite how long or how often they have been shopping with Greenfield Local Hub using the solution. This is not what a

		loyalty system is like in other website, therefore in the future the discount will need to be limited to how much a customer shops to be able to use the discounts.
Users able to change the application according to their accessibility needs	No	Because there is no functionality added to change the solution, such as dark mode, high contrast, or font size adjustment. This is crucial to allow a greater of users be able to use the services. Therefore, in the future this accessibility needs will be made to allow as many users as possible to be able to use the solution comfortably. Feedback on accessibility 15. Would you like a dark mode for the website? More details  17. What would you rate overall for the accessibility of the website? (1 Star = Horrible, 5 Stars = Perfect) More details 
Add access control pages based on user roles	Yes	Since users cannot access page that they are not supposed to have, this access control is also added to different endpoints to avoid bad actors from having access to sensitive data though the endpoints. 39. Is all the endpoints in all controllers configured to be secure from unauthorised users? More details  In the future, functionalities that are fixed or complete will have their authorisation for the appropriate users, so these functionalities can be used.

Non-Functional Requirements

Performance

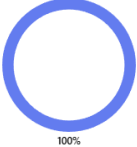
Requirement	KPI	Met	Explanation and Future Improvement
The webpages must load in industry standard time.	Webpages containing just text should load within 15 milliseconds. Webpages containing less than	Not fully	This is because webpages load within 15 milliseconds like the homepage or the products' page as they need to fetch a small amount of data. This does not impact the solution a lot currently as it does not need to load a lot of data, but this can be an issue in the future when it will be loading hundreds or products and basket items to users. This could

	20 high quality images should load within 30 milliseconds. Webpages containing more than 30 high quality images should load within 50 milliseconds.		be changed by either caching the products or basket items data in the servers or reducing the amount of data needed to show on the pages.
The system must handle a specific number of requests without degradation.	On average the system should handle 30 requests per second without degradation.	Yes	This is because the system can easily handle 30 requests per second and there is only a little amount of degradation which is in the expected amount.
CPU usage should not exceed a set amount	CPU usage should be under 20% on a S20 phone. CPU usage should be under 15% on iPhone15 Pro Max.	Not fully	This is because the website loads a lot of data inefficiently which makes the CPU usage skyrocket. This can be fixed by either compressing the data or loading the data in chunks to lower the CPU usage.
System should operate correctly under a multiplied expected load.	System should operate correctly under 4 times expected load.	No	This is because the data system is not designed to handle more load than expected. This means the system need to be changed to account for these factors. If it is not changed then it can impact the company negatively in the future with an actual commercial solution.
Input delay should be at or under industry standards	Input delay should be at or under 200 milliseconds when interacting with functions that process more than 100kb of data. Input delay should be at or under 100 milliseconds when interacting with functions that process more than 10kb of data.	Yes	This is because there are no complex calculations happening that are connected to inputs of the client, which ensures there is little input delay. But it could become an issue in the future if the data fetched from the server is not handled properly.
Payment should be processed quickly.	The payment processing speed should be at an industry standard.	No	This is because no payment processors have been added yet, which means this will need to be retested with a future improved solution that includes a payment processor.
Orders' status should be	Orders' status being updated under 2 minutes after the	Yes	This is because the changes are saved instantly to the server, which means whenever the user views the details of an order or products in the basket it will be up to date.

updated fast when changed.	status has been changed, during 50 or more orders' statuses are being changed.		
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Security

Requirement	KPI	Met	Explanation and Future Improvement
All data should be secured using industry standard encryption.	Data should be saved using SHA256 encryption.	Not fully	This is because a different encryption method is being used that is not known. This can result in malicious users able to have access to other users' data, which means the client will be fined if this ever happens in the future. To ensure this does not happen the encryption technique will need to be checked and changed accordingly.
Account password should be considered complex by industry standards	Account password should be at least 8 character long, contain letters, numbers and special characters to be consider complex.	Yes	This because all passwords are hashed in the database, which means attackers will have a harder time at getting access to a password of a user. This could be improved by adding other security options to the account, such as 2FA with the user's email or phone number.
Data should be encrypted during transit.	Website should enforce HTTPS on data when it is in transits. Data being transited should not contain sensitive information that is not encrypted before being transited.	Not fully	This is because the solution has not been configured to use 'HTTPS' when running. This can be bad in the future as this will expose potentially sensitive data to malicious users. To ensure this does not happen in the future, 'HTTPS' should be used for the domain to run this website with encrypted data transit.
There should be explicit user consent for cookies	Cookies should be visible on any page of the website with a clear way to accept or deny the cookies.	No	This is because a cookies section has not been created in webpages and the logic for the cookies has not been created. Which can lead to fines for the company, to ensure this does not happen in a future solution it will include a working and compliant cookies section.
Account details of the user should be censored.	Important details like passwords and emails will be replaced with asterisks or dots, these will be the same amount regardless the number of characters.	No	This is because users can view their details fully and their information has not been censored. This can be an issue in the future, if bad actors can view what users' details are from profile page.

Payment details should not be shown on the screen by default.	Card numbers will be replaced with asterisks or dots, when they are being typed in or just being displayed. Card holder's full name should be abbreviated to only show the surname on screen.	No	This is because a payment process has not been added and therefore no details are being taken from the users. But this will change in the future when a payment process is added to the solution and therefore the future solution will need to meet this requirement.
Users given specific roles	Users are only able to view pages that they are allowed based on their roles.	Yes	This is because users cannot access pages or endpoints that they are not authorised access. 39. Is all the endpoints in all controllers configured to be secure from unauthorised users? More details 

Capacity

Requirement	KPI	Met	Explanation and Future Improvement
The website should be able to handle a max number of users without the system crashing	The website should be able to handle max of 1000 users at the same time without the system crashing.	No	Since no systems has been added to spread users across servers to decrease likelihood of the server or system crashing.
The website should be able to handle many transactions per second under peak load	500 transactions per second should be sustained with less than or equal to 5% error rate.	No	Since no systems has been added to spread transactions across servers to decrease likelihood of the server or system crashing.
The website should be able to store data of many customers without overwriting previous data of other customers.	The website should be able to store up to 10TB of data for 10,000 customer accounts.	No	This is because the prototype uses the computer's storage, which it has been installed on and this inefficient. Therefore, in the future a storage server will be linked to the solution to store all data of users.

Scalability

Requirement	KPI	Met	Explanation and Future Improvement
New functions should be added at an industry standard pace to the application	On average, every new major function should be added within 5 to 10 days.	Yes	The current code and document are organised using the MVC pattern, which is made to allow developer to development solutions quickly. However, the code could be improved by separating it into modular

	On average, every new minor function should be added within 1 to 3 days.		functions as that will allow the code to be easier to manage to develop the solution further.
Traffic should be dispersed across different servers.	On average traffic should be dispersed evenly across all the different servers.	No	Since there is no system added to disperse traffic, therefore in the future a scalable system will need to be developed that will disperse traffic across different servers.
The code should be easily maintained and understandable by others.	A third party should be able to understand the coding conventions and logic being used in the solution and edit the code easily after a week of reviewing.	Not fully	This is because the current codebase has been commented to allow developers new to the solution to be able understand it. However, the codebase needs to be improved in the future by splitting redundant code into a separate function that is modular. This also will be applied to code sections that are too long or have too many nested pieces of code.

Usability and Accessibility

Requirement	KPI	Met	Explanation and Future Improvement						
Webpage should adapt to different device sizes	Webpage elements all are visible at different device sizes.	No	This is because the pages only look normal as intended on desktop screens with a specific size. Given this, in the future the CSS files need to be changed to allow webpages to fit all devices with different screen sizes. This problem is pointed from the user feedback (see below). 14. Why is all the CSS code used for the views NOT modular and robust for future views or pages? 1 Responses <table><thead><tr><th>ID ↑</th><th>Name</th><th>Responses</th></tr></thead><tbody><tr><td>1</td><td>anonymous</td><td>When making the solution tab smaller, the CSS code breaks and does not become modular at sizes smaller then 700px.</td></tr></tbody></table>	ID ↑	Name	Responses	1	anonymous	When making the solution tab smaller, the CSS code breaks and does not become modular at sizes smaller then 700px.
ID ↑	Name	Responses							
1	anonymous	When making the solution tab smaller, the CSS code breaks and does not become modular at sizes smaller then 700px.							
Website should have intuitive and consistent navigation	Any page should be able to be reached within 3 clicks from the homepage.	Yes	Since users find the website easy to navigate as shown from the user feedback (see below). 83. Website Usability & Experience More details Strongly Dislike Dislike Neutral Agree Strongly agree The website was easy to navigate However, this could be improved by adding a 'Back' button to allow users go back to different pages, which improves user experience and allows to be at a level of other similar websites.						
Interactive elements should provide immediate visual feedback when interacted with	All interactive components should change their state (colour, animation, size and etc) within 1 second of user interaction.	Yes	This is because all elements that can be interacted with have some sort of visual state change, such as buttons' colours changing slightly. But this can be improved in the future by adding a slight animation to indicate interactive elements more concretely and support users who have colour-blindness. In turn this improves the websites overall accessibility and use experience.						

Should have maintain consistent visual design and content tone throughout all pages	All pages should fully adhere to the design document of the website.	No	Since the users pointed out that the colour scheme did not fit the website or felt 'plain' and the User feedback is below. 8. Explain why the colours do NOT remind you of farming or fresh food? 2 Responses <table><thead><tr><th>ID ↑</th><th>Name</th><th>Responses</th></tr></thead><tbody><tr><td>1</td><td>anonymous</td><td>Its quite beige/white, the colour scheme only reminds me of an ecommerce website, but not something that would be related to food.</td></tr><tr><td>2</td><td>anonymous</td><td>its a bit plain</td></tr></tbody></table> To improve on this, in the future a background texture will be added to ensure the background does not seem 'plain', and a new colour scheme will be chosen to fit the locally produced food and farming vibe.	ID ↑	Name	Responses	1	anonymous	Its quite beige/white, the colour scheme only reminds me of an ecommerce website, but not something that would be related to food.	2	anonymous	its a bit plain
ID ↑	Name	Responses										
1	anonymous	Its quite beige/white, the colour scheme only reminds me of an ecommerce website, but not something that would be related to food.										
2	anonymous	its a bit plain										
The solution must show errors in an informative and understandable way.	Users should not have no complaints of them being confused or frustrated with errors.	Not fully	This is because many fields for forms have validation made from the default generated by the ASP.NET framework, but this means there are no validation built for special fields like an address. To handle this, in the future custom validation will be added ensure users cannot added a false address to an order, or input an order date in the past, and other types of validation is implemented with their respective informative error messages for the users.									

Prototype Future Improvements

1. Add other external services to allow to make an account with, such as a Microsoft account. This will allow more users to be able to use the solution, which means this will increase the chances of users making an account and using it to purchase products from the website. Furthermore, this enhances the user experience and that allows the solution stand out from the competition.
2. Modify the loyalty system to only allow customers to use discounts if they have hit a certain threshold of orders or money spent with the Greenfield Local Hub solution. By doing this, the loyalty system will be similar to other websites, which will allow the user to use it more easily and understand it faster. This will benefit Greenfield Local Hub as they will be able to convey the value of their website/solution to users effectively.
3. Allow the webpages to fit all screen size from different devices types. By doing this the website will be able to be used by a greater number of users, and that will increase the number customers ordering products from the website.
4. Adding improved validation for all form fields on the solution with appropriate error messages. This will reduce and remove invalid forms being submitted to the system. Furthermore, implementing this will decrease user frustration as they will be able to identify what errors they have made and fix them easily, due to the error messages that will be given to the user.
5. Allow the basket to be reachable on all pages for users by using the basket icon. This is will be developed in the future as it was requested by the users (see below).

65. If you were continuing this project, what would be your top priority be to improve the codebase?

[More details](#)

This will benefit GLH as customers will be able to find their basket from any page and increase the chance of them purchasing products using the Greenfield Local Hub website.